3 marks: Control abstraction
+ backtracking
+ BB bounding & branching

MODULE 3

Backtracking Technique

~~Br~~ Backtracking is an algorithm design technique that often makes it ~~to~~ possible to solve a large instance of combinational problems. The backtracking procedure works by adding the elements to the solutions ~~at~~ set, one element at a time. The selection of choice depends on the implicit criteria function that must be satisfied by the selected element. Searching with each choice (element) leads to a new set of choices, some of which may result into the solution of the problem.

If any of the choices does not lead to a solution (dead end), then we can backtrack to the junction (to previous step) and try the next choice. So backtracking allows us to backtrack one step up and start searching with another choice.

If no candidate is found then there is no meaning to extend the current partial solution, and if after trying all choices we do not find a solution, it means that the solution does not exist. So in the backtracking technique examine all the possible solutions of a program.

This technique requires 2 sets of constraints, they are:

(i) Explicit Constraints
(ii) Implicit Constraints

Explicit Constraints

These are the rules that restrict each component exact of the solution to take value of solution from a given set of values S.

The explicit constraints depends on the sub problem being solves and the tuples that satisfy the constraints define a possible solution corresponding to that subproblem.

Implicit Constraint

These are the rules that describes the ways in which the variable x_i's must relate to each other or which of the components of the solution satisfy the criteria function.

Notes

(i) Solution Space: It is the set of all tuples that satisfy the explicit constraints.

(ii) State space tree: The solution space is organized ~~in~~ as a ~~there~~ tree is called "state space tree".

(iii) Bonding function or criteria: It is a function created that used to kill the live nodes without generating all its children.

(iv) Live node: It is the node that has ~~must~~ be generated but none of its descendants are yet ~~to be~~ generated.

(v) Extended node (E-node):

It is the live node whose children are currently being generated.

(vi) Dead node : A node that is ~~not~~ not to be extended ~~for~~ further or all of its children have already been generated.

(vii) Answer node : It is the node that represents the answer of the problem that means the node at which the criteria is functions are maximize, minimize or satisfied.

(viii) Solution node : It is the node that has the ~~possib~~ possibility to become the answer node.

Example for backtracking technique based problem

(1) N - Queens Problem.

(2) Sum of subset problem

(3) Graph coloring problem

(4) Puzzles problem

(5) ~~Hamit~~ Hamiltonian cycle problem

N - Queens Problem

Place 4 Queues on the chess board based on the implicit and explicit conditions.

Conditions :

- Explicit constraints :-

If chess board consist of n-rows and n-columns, the rows and columns position for each queen must be taken from the set $S = \{1, 2, 3, \ldots, N\}$.

If some Queen is placed $\{1,2\}$ then that queen is actually placed on row 1 and column 2.

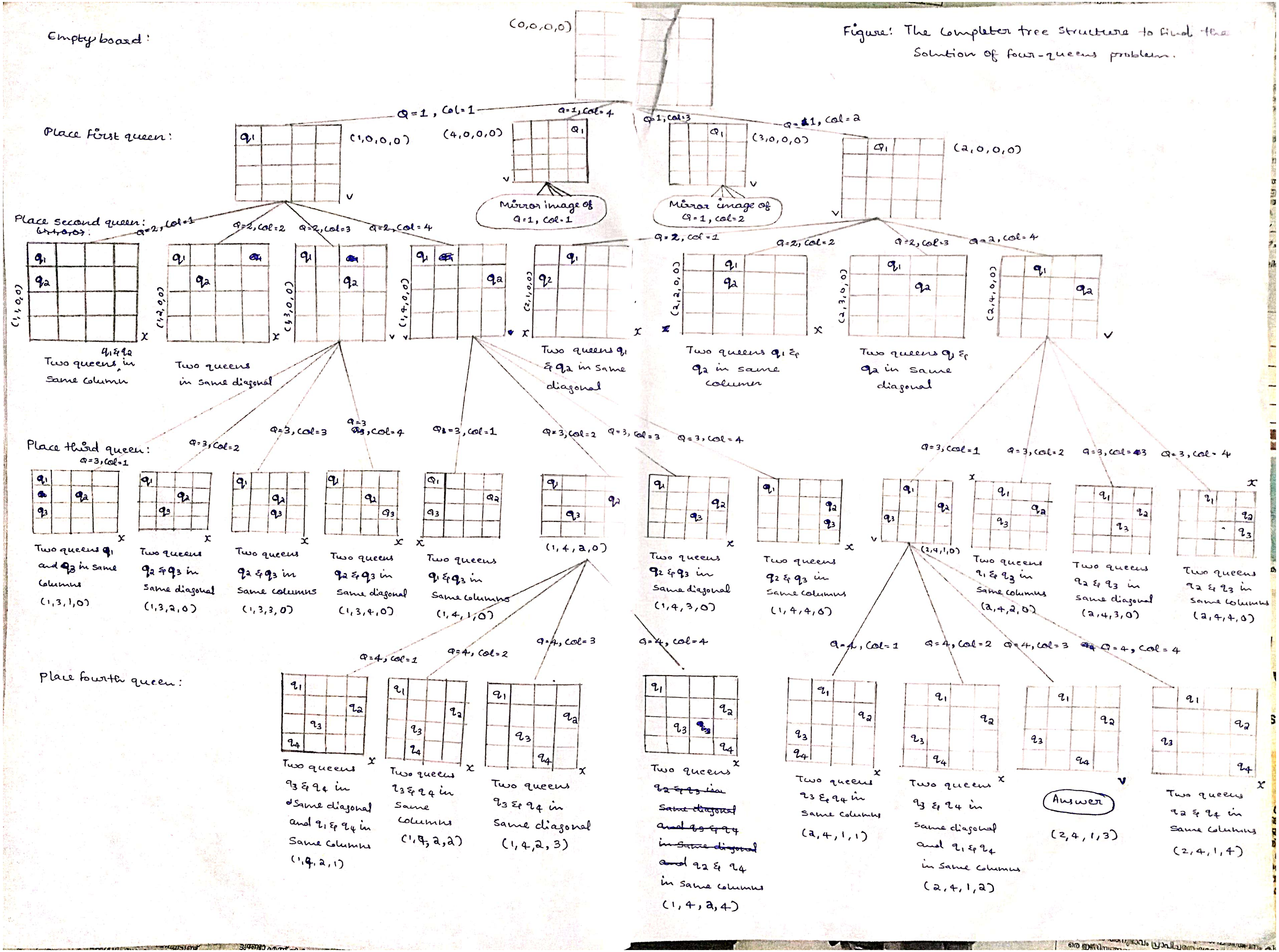
- <u>Implicit Constraint</u>

Implicit constraint are as follows:- → No two queens should be placed in the same row.

→ No two queens should be placed in the same column.

→ No two ~~queen~~ queens should be placed in the same diagonal.

Figure: The complete tree structure to find the Solution of four-queens problem.



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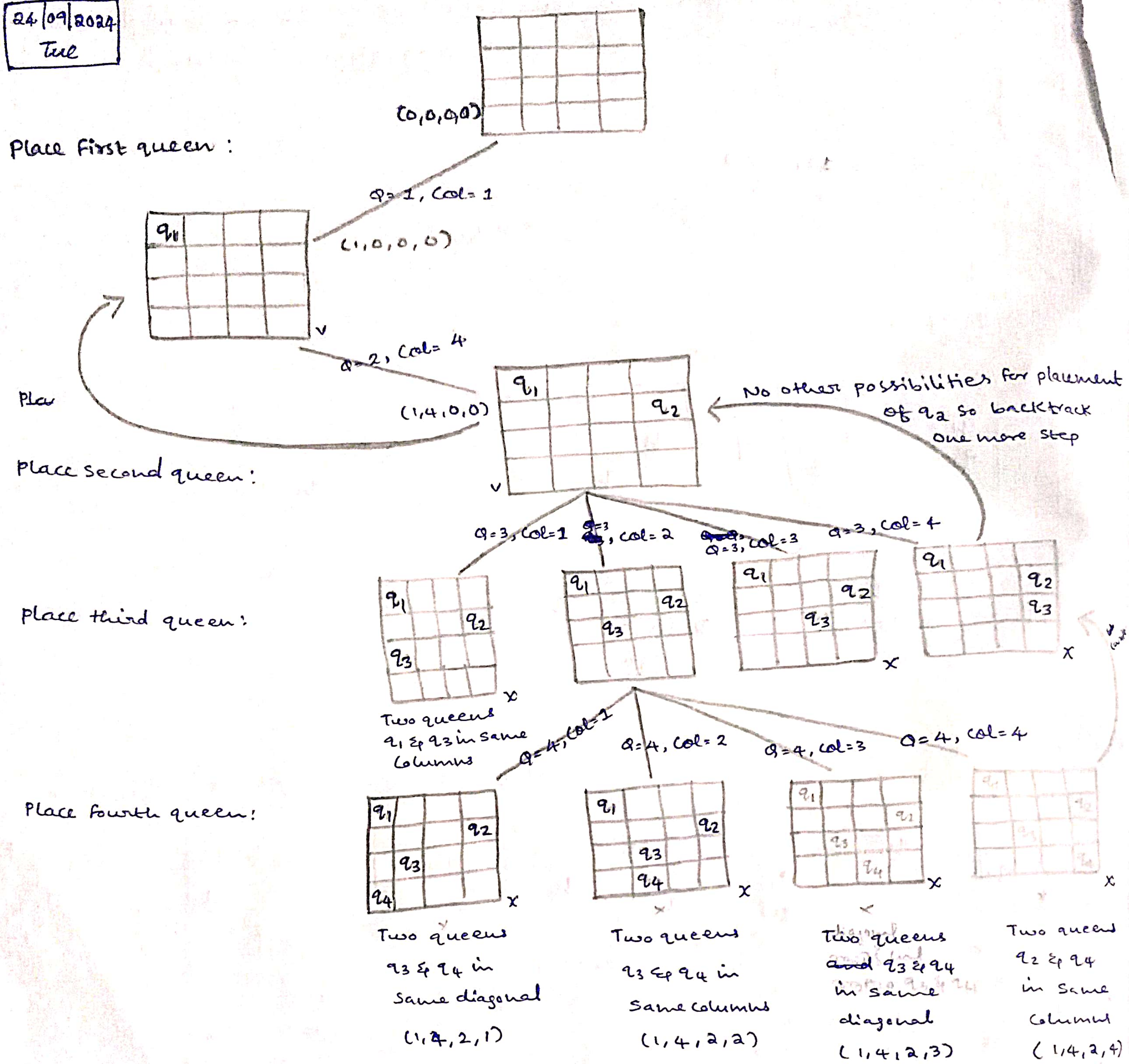

$q_1 = 1, \quad q_2 = 2, \quad q_3 = 3, \quad q_4 = 4$

$$S_1 \Rightarrow \begin{array}{cccc} q_1 & q_2 & q_3 & q_4 \\ (2, & 4, & 1, & 3) \end{array}$$

$$S_2 \Rightarrow (3, 1, 4, 2)$$

1st row 3rd col

4th row 2nd col

(write):

Other placement of q_3 also results in dead ends as shown in dotted edges so backtrack one more step

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Sum of subset problem

The sum of subset problem is defined as finding all the subsets 'S_i' of a given set of integers 'S' whose sum is added upto the given target 'M'.

If the target value M is 0, then we can immediately written true that is the elements of an empty set add upto zero.

If the given set S is empty and M less than zero then ~~x~~ it returns false.

The state space tree is constructed with the first value of the set S ~~calledwith~~ called the root of the tree. Here, we use the backtracking technique. The root of the tree represents the starting point with no decisions yet ~~to~~ taken on any input. The left child of the node indicates the inclusion of the selected element from the set S and the right side of the tree indicates the exclusion of the selected element from the set S. There is a subset 'X' of S that sums to M if and only if one of the following statements is true.

→ There is a subset of X that ~~itdic~~ includes X and whose sum is M.

→ There is a subset of X that excludes X and whose sum is M.

→ The value of the element ~~X-size~~ x_i's of the solution is either 1 or 0 depending on ~~the~~ the whether the weigh w_i of element x_i is included or not included respectively.

Question (IMP) (Essay)

Consider a set $S = \{2,3,5,6,7,9,10\}$ and $M=15$, Solve it for obtaining the sum of subsets using backtracking technique?

Solution

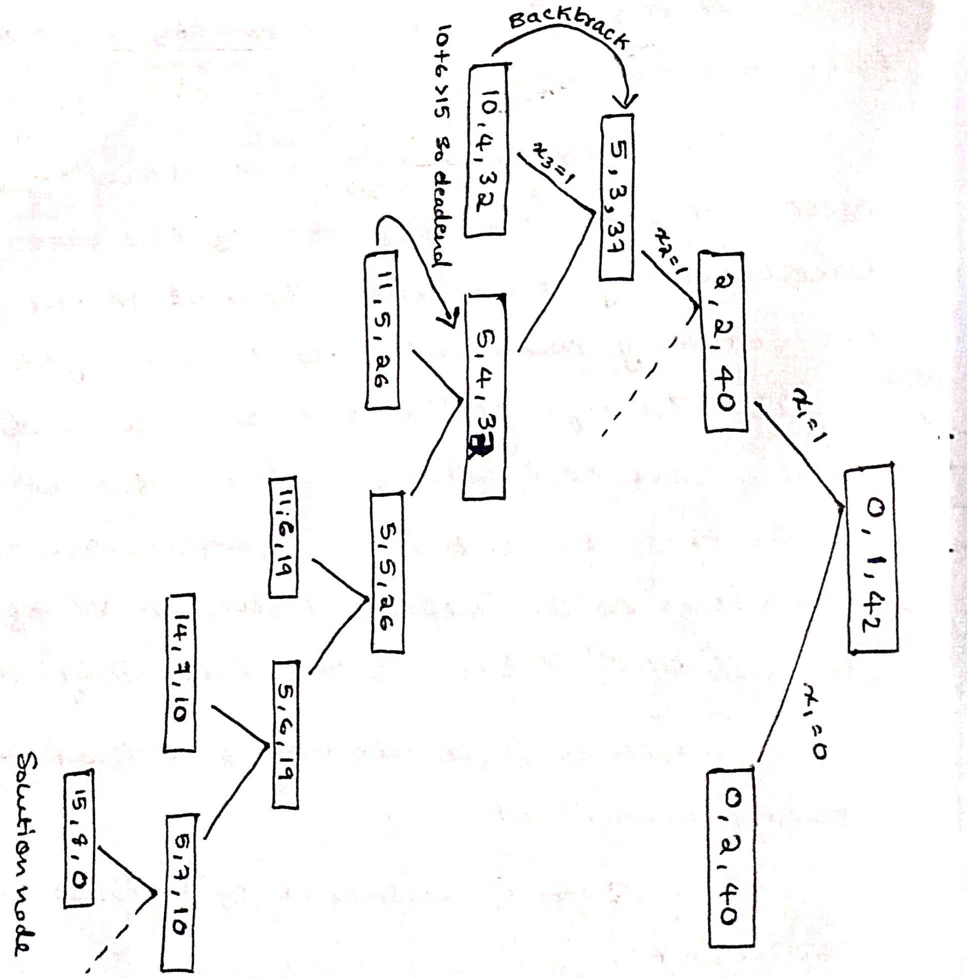
Sumsofar,	Iteration,	remainingweight
0	1	42

$S = \{2, 3, 5, 6, 7, 9, 10\}$

$x_i \rightarrow$ x_1 x_2 x_3 x_4 x_5 x_6 x_7

Inclusion represented by 1 and exclusion represented by 0.

$S_1 =$



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B & B (Branch & Bound) (Essay - 6 marks)

Solve 8 puzzle problem using Branch & Bound Technique

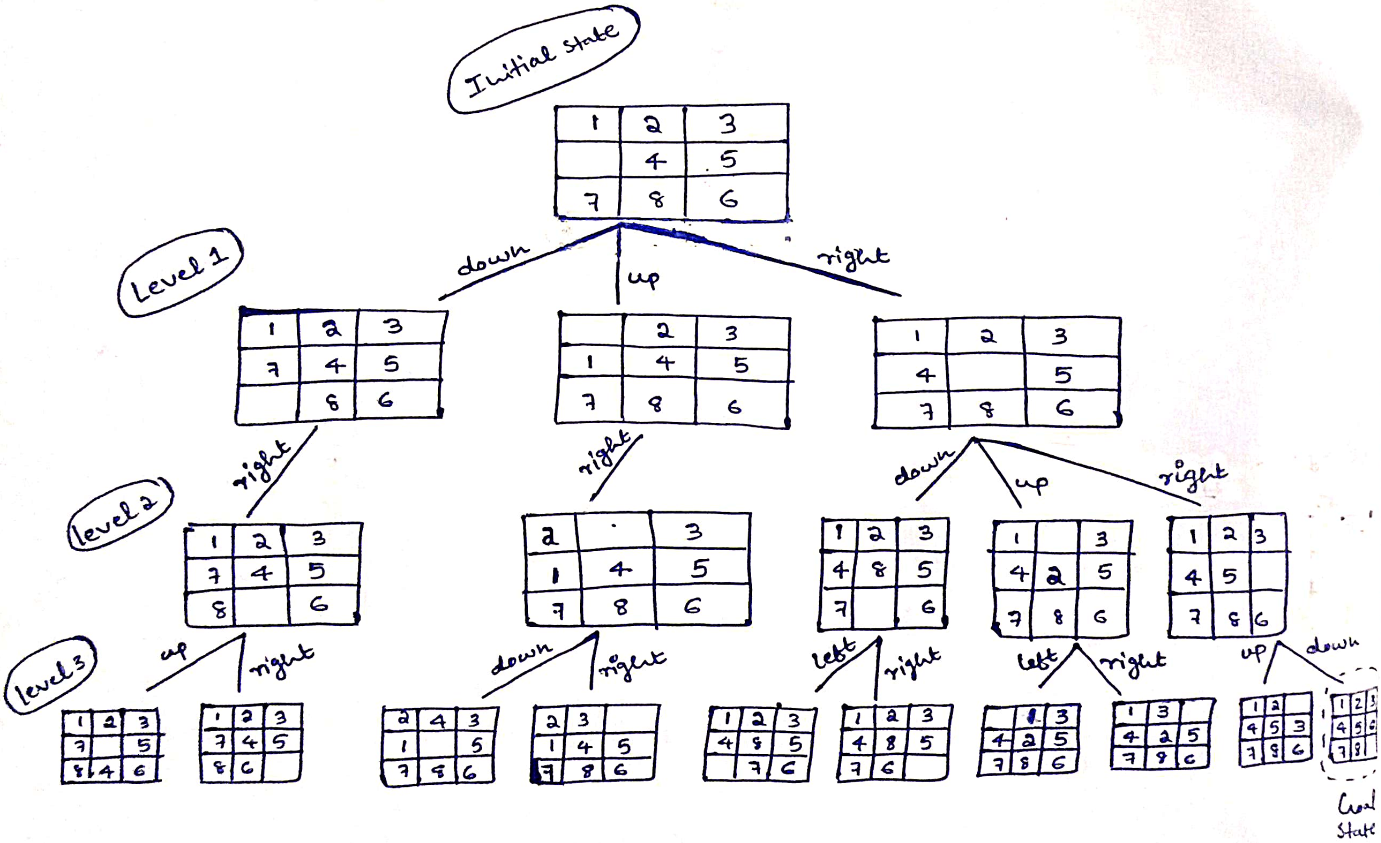
The initial state of the node is

Initial State

1	2	3
	4	5
7	8	6

Goal State

1	2	3
4	5	6
7	8	



Question

Solve 8 puzzle problem using Branch and Bound Technique

Initial State

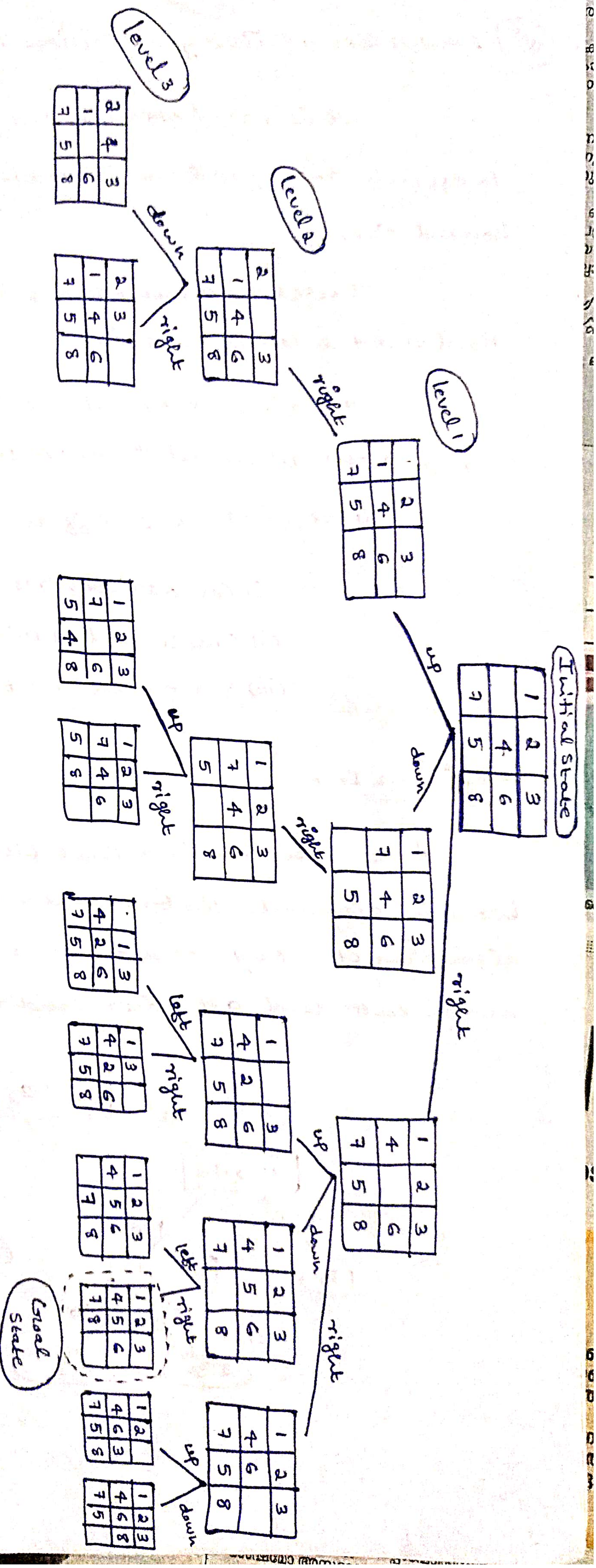
1	2	3
	4	6
7	5	8

Goal State

1	2	3
4	5	6
7	8	

Solution

In third level, right movement.



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6 mark
Essay

IMP

Lower Bound Theory (Decision Tree)

It is based ~~on~~ upon the calculation of minimal time that is required to execute an algorithm is known as lower bound theory or base bound theory.

It uses ~~the~~ a number of users methods or techniques to find out the lower bound.

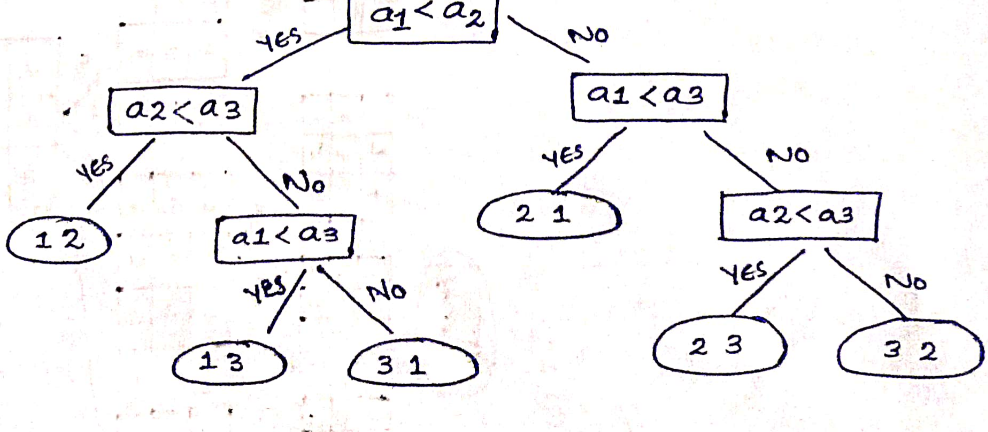
Its aim is to calculate a ~~mini~~ minimum number of comparisons required to execute an algorithm.

Here, its use different techniques. They are:

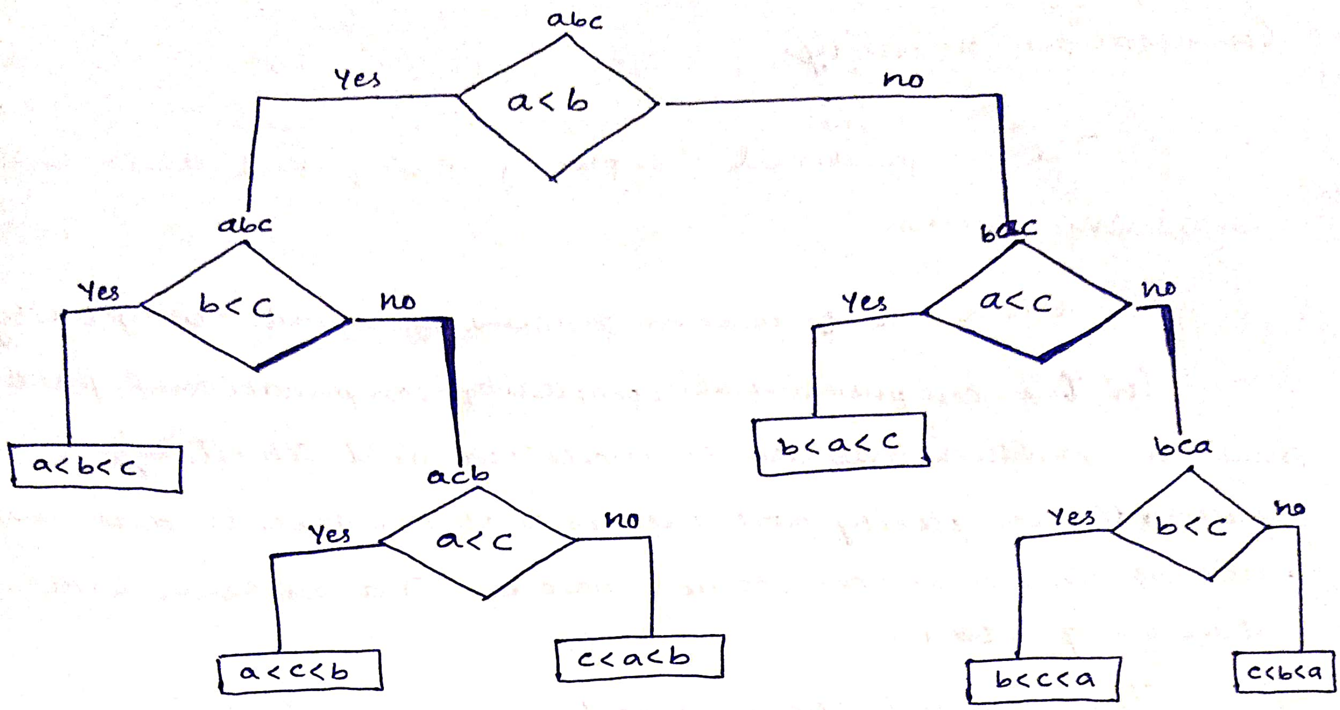
(i) Comparison trees
(ii) Oracle and advisory arguments
(iii) State space method.

Decision Tree

A decision tree is a full binary tree that shows the comparison between elements that are executed by an appropriate sorting algorithm operating on an input of an given size. Control, data movement and all other conditions of the algorithm are ignored.



Decision tree for 3-element insertion sort



Decision tree – A convenient model of algorithms involving comparisons in which:-

- Internal nodes represent comparisons
- Leaves represent outcomes

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